



Problem solving with non-linear relationships

1 Which of these are triangular numbers? Tick **2** correct answers

☐ 0☐ 1☐ 5☐ 10☐ 20

2 Use a calculator to work out the value of 3^8 . _____. Fill in the blank

3 The next term in the geometric sequence 0.4, 2, 10, ... is _____. Fill in the blank

4 Which of these could be the first 4 terms in an arithmetic sequence? Tick **2** correct answers

☐ 28, 22, 16, 11, ...☐ 136, 163, 190, 209, ...☐ 13, 49, 85, 121, ...☐ 17, 9, 1, -7, ...☐ -21, -8, 8, 21, ...

5 What is the n^{th} term rule for the linear sequence which starts 13, 8, 3, -2, ...? Tick **1** correct answer

☐ $18 - 5n$ ☐ $13 - 5n$ ☐ $8 - 5n$ ☐ $5n + 8$ ☐ $5n + 13$ 

6 Which of these could be the first 5 terms of a sequence with a common second difference? (These are called quadratic sequences). Tick **3** correct answers

☐ 2, 3, 5, 8, 12, ...

☐ 8, 12, 18, 28, 40, ...

☐ 9, 10, 12, 16, 24, ...

☐ 12, 17, 27, 42, 62, ...

☐ 15, 21, 28, 36, 45, ...

